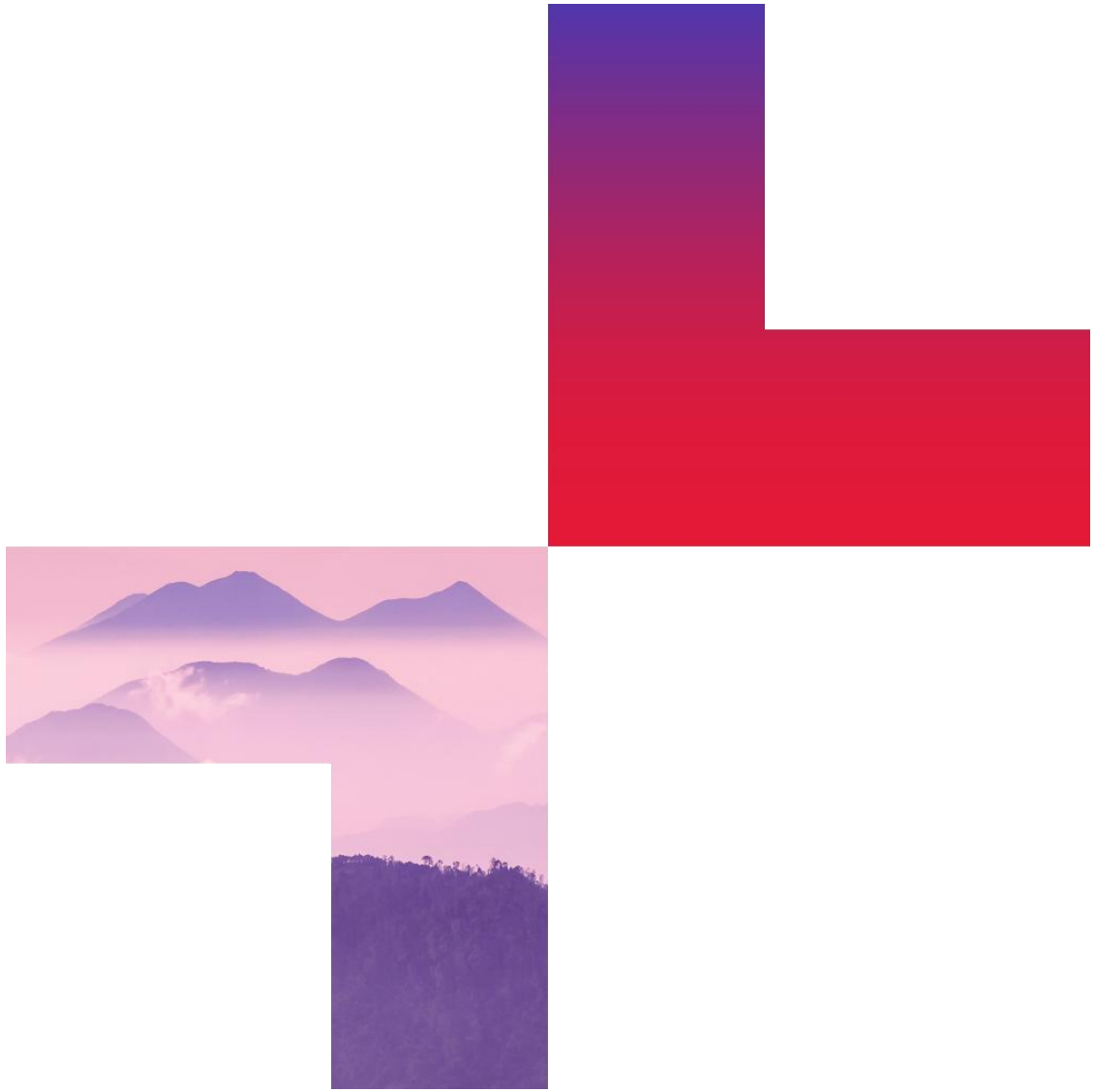
Texas MISA Fall 2026 Case Competition



November 2026

CGI

Costco Workforce Planning Challenge

About Costco

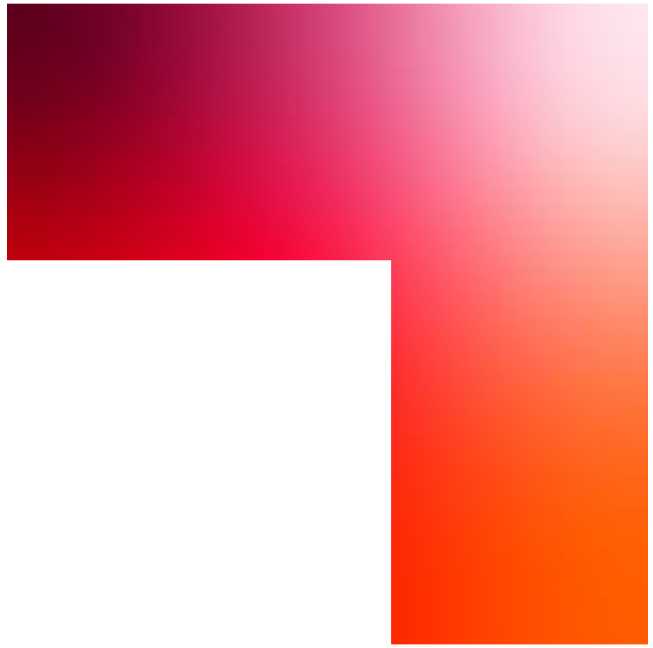
Costco Wholesale Corporation operates membership warehouses and e-commerce sites built around:

- Low prices
- High sales volume
- Rapid inventory turnover
- A limited selection of nationally branded and private-label products
- Multiple specialty departments and member services

Because of this operating model, staffing decisions affect more than checkout speed. Labor levels also influence product availability, receiving, replenishment, pharmacy services, membership support, safety, and the overall member experience.

© 2026 CGI Inc.

The Business Challenge



Warehouse leaders make staffing decisions in an environment where demand can change based on:

- Day of the week
- Customer traffic
- Holidays and promotions
- Weather
- Warehouse size
- Department activity
- Employee availability and qualifications
- Callouts and overtime requirements

Too much labor increases operating costs. Too little labor can result in longer wait times, slower replenishment, reduced product availability, and lower member satisfaction.

Costco needs a practical way to balance labor, staffing coverage, and service levels.

© 2026 CGI Inc.

The Case Prompt

Staffing levels must align with customer demand while controlling labor costs. Design a solution that helps warehouse leaders forecast staffing needs, optimize schedules, and maintain service levels.

Your Solution Should Help a Warehouse Leader Understand:

- When demand is expected to increase or decrease
- How staffing needs vary by department and time period
- How schedules or labor coverage should be adjusted
- How the recommendation affects labor costs and service levels
- What actions should be taken when demand or staffing conditions change

© 2026 CGI Inc.

The Decision Costco Needs to Make

Your team must help Costco answer:

How should the warehouse allocate labor across departments and time periods to meet expected demand at the lowest practical cost while maintaining service levels?

Your plan must cover all three scenarios in the planning week (Feb 12-18, 2026):

- **Baseline:** the expected week, including Valentine's Day and Presidents' Day
- **Peak Demand:** a promotion launch and a winter-storm advisory that drive extra traffic
- **Disruption:** illness and crew shortages that reduce available staff

Your recommendation should show where, when, and how much labor is needed, which assumptions drive it, and what the warehouse leader should do when conditions shift.

© 2026 CGI Inc.

Solution Expectations

Build a proof of concept, not just a dashboard. Any working tool a warehouse leader could use qualifies.

Your solution should demonstrate how Costco could use data to make better staffing decisions.

A dashboard may be included as part of the solution, but it should not be the solution by itself. Teams should go beyond displaying metrics and demonstrate how their approach:

- Forecasts or estimates future demand
- Converts demand into staffing requirements
- Recommend schedules, staffing levels, or labor coverage
- Balances labor cost with service-level objectives
- Responds to changing conditions or scenarios
- Explains why a recommendation was made

© 2026 CGI Inc.

Available Data and Assumptions

Available Data

Teams will receive a simulated Costco workforce-planning dataset containing:

- Three Costco warehouses with different sizes and operating characteristics
- Three departments: Front End, Merchandising/Stocking, and Pharmacy
- Six weeks of historical demand and operating data (Jan 1-Feb 11, 2026)
- Weekday, weekend, and holiday patterns
- Regular and overtime labor-cost assumptions
- A one week planning period (February 12-18, 2026) to forecast staff

Case Assumptions

- The data is simulated for competition purposes and does not represent confidential Costco data
- Students should clearly identify any additional assumptions they make
- Labor cost is calculated the same way for every team: regular staff first, then cross-trained staff, then overtime. Rates are provided in the data.
- Teams must submit staffing plans for all three scenarios: Baseline, Peak Demand, and Disruption
- Service-level measures should be used to evaluate whether a staffing recommendation meets operational expectations

© 2026 CGI Inc.

What to Submit

Presentation

Include your diagnosis, forecast, recommendation, and how it changes across the three scenarios, including a short demo of your proof of concept (can be live or recorded)

Staffing Plan

The completed TEAM_SUBMISSION_TEMPLATE.csv, with the total employees for every warehouse, department, date, and shift in all three scenarios. You don’t need to split totals into regular, overtime, and cross-trained staff.

Proof of Concept

A working tool that takes a scenario as input and shows staffing, labor cost, and service risk.

Supporting Analysis

Include appendix slides or files showing your forecast method, key assumptions, and cost and service compared with the current draft schedules.

© 2026 CGI Inc.